

# Layered Oil-Price Pass-Through in India: Evidence from Wholesale, Retail Fuel, and Consumer Prices

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## Abstract

This paper studies how global oil-price shocks travel through India's domestic price system. Rather than estimate only oil price effect on headline CPI, it tracks shock across wholesale, retail fuel, fuel-sensitive consumer and headline consumer price layers. Monthly log-difference ADL models are estimated by Newey-West HAC inference with bootstrap symmetry checks. Results show layered attenuation. Retail petrol responds strongly to Brent shocks. The preferred post-2010 WPI Fuel and Power model responds strongly to rupee oil shocks and passes the diagnostic gate. CPI Fuel and Light shows smaller bridge evidence and headline WPI shows moderate pass-through. Pass-through is weak and statistically insignificant at the headline CPI endpoint. Except for a marginal retail petrol result, strong short-run asymmetry is not supported. In general, oil shocks are still visible in fuel layers but weaken in broad consumer inflation.

**Keywords:** Oil-price pass-through; inflation; India; wholesale price index; consumer price index; retail fuel prices.

**JEL codes:** C22; E31; Q43; F31.

## 1. Introduction

India imports much of its crude oil and prices that crude in United States dollars. Domestic oil pressure therefore depends on the Brent price and on the rupee-dollar exchange rate, so a good measure of the domestic oil shock is the rupee-denominated oil price,  $\text{oil}^{\text{INR}}_t = \text{Brent}^{\text{USD}}_t \times \text{INR}_t/\text{USD}_t$ . If Brent rises or the rupee weakens, the rupee oil price rises, and the cost pressure should be strongest in the layers closest to fuel and then, with attenuation, in broader price aggregates (Mandal et al., 2012; Bhanumurthy et al., 2012).

There's a familiar puzzle in this transmission. Petrol and diesel prices closely follow crude, and wholesale fuel inflation moves with rupee oil prices, but headline consumer inflation does not move one-for-one with oil. The Wholesale Price Index (WPI) tracks producer cost pressure more closely than the Consumer Price Index (CPI), which is the household-facing inflation index and is weighted heavily towards food, housing, services, health and education. Because fuel is only part of the CPI basket, even a severe oil shock can be diluted by the time it reaches headline CPI inflation.

Most existing studies of Indian oil pass-through estimate one equation linking oil to either WPI or CPI inflation (Mandal et al., 2012; Bhanumurthy et al., 2012; Pradeep, 2022). That approach gives one elasticity, but does not show where the signal weakens. The contribution of this paper is to treat oil-price pass-through as a layered price-system problem rather than a single elasticity. The chain studied here moves from rupee oil price to wholesale prices, from Brent to the Petroleum Planning and Analysis Cell (PPAC) Delhi retail petrol price, from PPAC retail petrol to CPI Fuel and Light, and from rupee oil to headline CPI. Each layer is estimated as a short-run asymmetric autoregressive distributed lag (ADL) model in monthly log differences.

The institutional background is also relevant. Domestic fuel prices were not always fully linked to international prices, and reform episodes changed the way international oil movements entered petrol and diesel prices. Pradeep (2022) studies this reform setting and

places asymmetry at the centre of the question. This paper uses that background more narrowly. It does not claim that deregulation caused the post-2010 estimates. It treats the reform period as context for reading stronger wholesale and retail fuel pass-through once pricing became market-linked.

The WPI versus CPI distinction is useful here because the two indices measure different layers of the price system. WPI more closely tracks producer and wholesale cost pressure. CPI is the consumer-facing aggregate used for inflation targeting, but it contains many items whose monthly movements need not follow oil prices. A layered design is therefore more cautious than a single headline CPI equation. If oil is strong in fuel layers but weak in headline CPI, then attenuation is the right conclusion, not irrelevance.

For inference, Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors (Newey & West, 1987) are used, and the cumulative pass-through statistics  $CPT^+$  and  $CPT^-$  summarise the response to positive and negative shock terms. Bootstrap resampling gives a robustness check for asymmetry. Diagnostic tests include serial correlation, functional form and recursive stability, and a model-gate rule distinguishes main, caveated and excluded equations. Bai and Perron (2003) structural-break logic and the unit-root test battery, including KPSS checks (Kwiatkowski et al., 1992), are used to assess the differenced specification.

The estimates suggest layered attenuation. Retail petrol shows the strongest direct consumer-side fuel response, with  $CPT^+ = 0.3459$  ( $p < 0.001$ ). The preferred post-2010 WPI Fuel and Power model gives  $CPT^+ = 0.5205$  ( $p < 0.001$ ) and passes the diagnostic gate. CPI Fuel and Light, the consumer fuel bridge, shows  $CPT^+ = 0.1777$  ( $p = 0.0021$ ). Headline WPI reports a statistically significant but modest  $CPT^+ = 0.0301$  ( $p = 0.0240$ ). Headline CPI shows  $CPT^+ = 0.0213$  with  $p = 0.1220$ , positive in sign but not statistically significant at conventional levels. A formal Wald test rejects equality between the retail petrol and headline CPI positive pass-through estimates at  $F = 14.3499$ ,  $p = 0.0002$ . Asymmetry is weak overall; only retail petrol shows a marginal  $CPT^+ = CPT^-$  rejection at the ten per cent level ( $p = 0.0999$ ).

The remainder of the paper is arranged as follows. Section 2 describes the data and variables. Section 3 details the methodology. Section 4 reports results across price layers. Section 5 deals with robustness, diagnostics and limitations. Section 6 discusses implications and concludes.

## **2. Data and variables**

This monthly dataset combines official Indian price series with international oil and exchange-rate data. Headline WPI and WPI Fuel and Power are from the Office of the Economic Adviser (OEA), chained to the 2011-12 base (Office of the Economic Adviser, 2017). The headline CPI source file is the OECD all-items India CPI series accessed via FRED (INDCPIALLMINMEI), and CPI Fuel and Light is the processed all-India MoSPI component series. Delhi retail petrol prices are from the Petroleum Planning and Analysis Cell (PPAC, 2024). Brent crude is from the World Bank Pink Sheet (World Bank, 2026). The CPI pipeline uses the FRED EXINUS file, and the WPI pipeline uses the updated FRED EXINUS<sub>latest</sub> file. All series are monthly.

The domestic oil shock is constructed as  $\text{oil}^{\text{INR}}_t = \text{Brent}^{\text{USD}}_t \times \text{INR}/\text{USD}_t$ , where  $\text{INR}/\text{USD}_t$  is the monthly average exchange rate. This rupee oil price captures both the Brent component and the exchange-rate component of cost pressure on Indian importers and refiners. For the retail petrol model, the regressor is the Brent shock directly, because domestic retail fuel prices are administered with reference to international product prices and the rupee component is partially absorbed by tax and pricing rules. For the CPI Fuel and Light bridge model, the regressor is the PPAC petrol shock, because that is the price the fuel-sensitive consumer layer faces.

The samples are deliberately allowed to differ by layer. The headline WPI series gives the longest window, from 1983-05 to 2026-03. WPI Fuel and Power begins later, in 1995-05. The PPAC Delhi petrol and headline CPI equations run from 2004-08 to 2024-12 and therefore give the main 20-year consumer-side window. CPI Fuel and Light begins only in 2011-05, so it is used as a bridge between retail fuel and consumer inflation rather than as a mandatory headline result. This is why the title does not state one common date range for every layer.

$$\Delta x_t = 100 \times (\ln(x_t) - \ln(x_{t-1})) \quad (1)$$

Equation (1) defines the monthly log difference of any series  $x_t$ . The factor of 100 means coefficients on  $\Delta x_t$  can be read approximately as percentage responses. The positive and negative shock components are defined as  $\Delta x^+_t = \max(\Delta x_t, 0)$  and  $\Delta x^-_t = \min(\Delta x_t, 0)$ . The negative-shock variable is non-positive by construction, so the cumulative pass-through CPT<sup>-</sup>, which is the sum of coefficients on lagged  $\Delta x^-$  terms, should not be interpreted as the response to a positive shock with the opposite sign.

**Table 1. Data, transformations, and model role.**

| Series                   | Source                         | Transformation         | Sample             | Role                    |
|--------------------------|--------------------------------|------------------------|--------------------|-------------------------|
| Brent crude price        | World Bank Pink Sheet monthly  | Monthly log difference | Matched to layer   | Global oil shock        |
| INR/USD exchange rate    | FRED EXINUS / EXINUS_latest    | Monthly log difference | Matched to layer   | Exchange-rate component |
| Rupee oil price          | Brent $\times$ FRED INR/USD    | Monthly log difference | Matched to layer   | Domestic oil shock      |
| Headline WPI             | OEA WPI files, base 2011-12    | Monthly log difference | 1983-05 to 2026-03 | Wholesale endpoint      |
| WPI Fuel and Power       | OEA WPI files, base 2011-12    | Monthly log difference | 1995-05 to 2026-03 | Wholesale fuel layer    |
| PPAC Delhi retail petrol | PPAC RSP pre/post-2017 files   | Monthly log difference | 2004-08 to 2024-12 | Retail fuel layer       |
| CPI Fuel and Light       | MoSPI all-India Fuel and Light | Monthly log difference | 2011-05 to 2024-12 | Consumer fuel bridge    |
| Headline CPI             | FRED/OECD INDCPIALLMINMEI      | Monthly log difference | 2004-08 to 2024-12 | Consumer endpoint       |

*Note: All transformations are multiplied by 100, so coefficients can be read approximately as percentage responses. Samples differ because the official series are available over different periods. Source: Author's compilation from cited official sources.*

### 3. Methodology

Each price layer is modelled with a short-run asymmetric ADL specification in monthly log differences. Let  $\Delta y_t$  denote the monthly log change of the layer-specific dependent variable and  $\Delta x_t$  denote the relevant shock variable. The estimating equation is

$$\Delta y_t = \alpha + \sum_{i=1}^p \varphi_i \Delta y_{t-i} + \sum_{j=0}^q \beta_j^+ \Delta x_{t-j}^+ + \sum_{j=0}^q \beta_j^- \Delta x_{t-j}^- + \gamma' Z_t + \mu_m + \varepsilon_t \quad (2)$$

Here  $\Delta y_t$  is monthly inflation in the relevant price index and  $\Delta x_t$  is the layer-appropriate shock. The lag choices are model-specific and are fixed before interpreting the results. In the CPI pipeline, the autoregressive lag is  $p = 3$ , selected by AIC over  $p = 1$  to 4, and the oil-shock lag window is  $q = 3$ . In the WPI pipeline, the headline and Fuel and Power models use AR(12) terms and oil-shock lags 0 to 6. The vector  $Z_t$  contains only the controls used in each estimated equation, and  $\mu_m$  denotes month fixed effects where they are included. The innovation term  $\varepsilon_t$  is allowed to be heteroskedastic and serially correlated.

The exact specifications used for the main and contextual rows are: headline WPI, ADL(12,6) with month fixed effects; preferred WPI Fuel and Power, ADL(12,6) with exchange-rate controls and month fixed effects, estimated from April 2010 onward while excluding April to September 2020; PPAC retail petrol, ADL(3,3) with IIP growth, petrol and diesel deregulation dummies, a COVID dummy, and month fixed effects; PPAC petrol to CPI Fuel and Light, ADL(3,3) without extra controls; and headline CPI M1, ADL(3,3) with IIP growth, petrol and diesel deregulation dummies, a COVID dummy, and month fixed effects. Full lag coefficient outputs are retained in the replication material. The article reports cumulative pass-through and compact diagnostic summaries in the main text to keep the argument readable.

Variables across layers match the channel each layer represents. The headline WPI model employs rupee oil shocks as  $\Delta x_t$ . The WPI Fuel and Power model also uses rupee oil shocks. Brent shocks are used in the PPAC retail petrol model. PPAC petrol shocks are used in the CPI Fuel and Light bridge model. In the headline CPI model, rupee oil shocks are used. This matching keeps each regression interpretable as a pass-through equation for its layer.

Cumulative pass-through is summarised by  $CPT^+ = \sum_j \beta_j^+$  and  $CPT^- = \sum_j \beta_j^-$ . Hypothesis tests are reported for  $CPT^+ = 0$ ,  $CPT^- = 0$ , and  $CPT^+ = CPT^-$ . Inference uses Newey-West HAC standard errors with bandwidth  $\text{floor}(0.75N^{1/3})$  (Newey & West, 1987). A restricted-residual circular block bootstrap with 4,999 replications gives a robustness check on the asymmetry test. Predictive precedence between oil and the dependent variable is examined using Granger-style tests on lagged shock terms; these are reported as predictive precedence rather than structural causality. Tests for structural-break behaviour are performed using Bai and Perron (2003) tests, and KPSS tests confirm stationarity of log-differenced series (Kwiatkowski et al., 1992).

A model is deemed claim-bearing only if it passes a mandatory diagnostic gate, specifically a Breusch-Godfrey test for residual serial correlation, a HAC-robust RESET test for functional form, and a recursive cumulative sum test for parameter stability. Models that fail one of these checks are reported with an explicit caveat or excluded from the main claims. The full-sample pooled WPI Fuel and Power model fails this gate and is retained only

as transparency context. The preferred post-2010 excluding-COVID Fuel and Power specification passes the gate and carries the wholesale fuel mechanism claim. ADF and Phillips-Perron tests confirm the first-differenced design; KPSS passes the CPI differenced variables and most WPI differenced variables, while WPI inflation carries a KPSS break-related caveat handled through the Bai-Perron check. The framework is short-run by design. Since the interest is in monthly transmission rather than long-run equilibrium adjustment, there are no NARDL bounds tests, no error-correction terms, and no long-run pass-through claims.

## 4. Results

### 4.1 Cumulative pass-through across price layers

Table 2 reports the central estimates. The reading is layer by layer. The retail petrol model uses Brent shocks over 2004-08 to 2024-12. The wholesale models use rupee oil shocks over the longest available samples. The CPI Fuel and Light bridge uses PPAC petrol shocks over 2011-05 to 2024-12. The headline CPI model uses rupee oil shocks over 2004-08 to 2024-12. Sample sizes vary across layers because the official series are available over different periods, so the full five-layer ranking is descriptive rather than a single common-sample estimate.

**Table 2. Cumulative pass-through by price layer.**

| Layer               | Sample             | N   | CPT+   | p      | CPT-   | p      | Asym. p | Verdict                                    |
|---------------------|--------------------|-----|--------|--------|--------|--------|---------|--------------------------------------------|
| PPAC retail petrol  | 2004-08 to 2024-12 | 245 | 0.3459 | <0.001 | 0.1912 | 0.0002 | 0.0999  | Strong direct fuel; marginal asymmetry     |
| WPI Fuel and Power* | 2010-04 to 2026-03 | 186 | 0.5205 | <0.001 | 0.4195 | <0.001 | 0.1642  | Preferred wholesale fuel; diagnostics pass |
| CPI Fuel and Light  | 2011-05 to 2024-12 | 164 | 0.1777 | 0.0021 | 0.1058 | 0.1741 | 0.4554  | Bridge evidence; shorter sample            |
| Headline WPI        | 1983-05 to 2026-03 | 515 | 0.0301 | 0.0240 | 0.0374 | 0.0012 | 0.6727  | Modest but significant                     |
| Headline CPI        | 2004-08 to 2024-12 | 245 | 0.0213 | 0.1220 | 0.0006 | 0.9375 | 0.2408  | Weak; not significant                      |

*Note: CPT+ and CPT- are sums of positive- and negative-shock coefficients across the ADL lag window. P-values use Newey-West HAC inference. Asym. p is the p-value for the test  $CPT^+ = CPT^-$ . WPI models use rupee oil shocks; the retail petrol model uses Brent shocks; the CPI Fuel and Light row is the PPAC-to-Fuel bridge. The preferred WPI Fuel and Power row excludes April to September 2020. Samples differ by data availability, so the table is descriptive; the consumer-chain common-sample check is reported below. Source: Author's estimates from model outputs.*

### 4.2 Wholesale layer

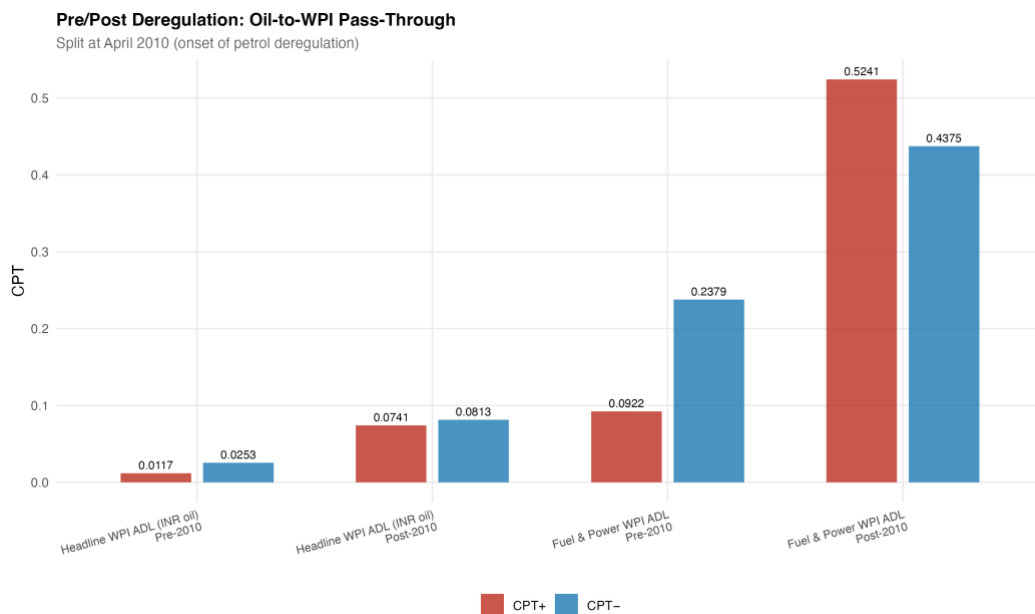
Headline WPI shows statistically significant but modest pass-through from rupee oil shocks. The combined response to positive shocks is  $CPT^+ = 0.0301$ ,  $p = 0.0240$ , and the cumulative response to negative shock terms is  $CPT^- = 0.0374$ ,  $p = 0.0012$ . The asymmetry test does not reject equality ( $p = 0.6727$ ), so the wholesale endpoint does not support short-run asymmetry. The adjusted  $R^2$  of 0.4207 over 515 observations suggests that lagged inflation, the oil shock and seasonal structure account for a large share of monthly headline

WPI variation. The model passes the Breusch-Godfrey, HAC-RESET and recursive CUSUM checks, so it is treated as claim-bearing.

The preferred WPI Fuel and Power model has a stronger oil signal, with  $CPT^+ = 0.5205$  ( $p < 0.001$ ) and  $CPT^- = 0.4195$  ( $p < 0.001$ ) over 186 observations. Point estimates indicate that the wholesale fuel layer responds strongly within the lag window. The asymmetry p-value of 0.1642 does not reject equality of positive and negative cumulative effects. The model clears the mandatory diagnostic gate: Breusch-Godfrey  $p = 0.0746$ , HAC-RESET  $p = 0.5443$  and recursive CUSUM  $p = 0.6999$ .

The older full-sample pooled Fuel and Power model still provides useful context but no longer carries the preferred inference. On 371 observations it returns  $CPT^+ = 0.2866$  and  $CPT^- = 0.2677$ , but fails the HAC-RESET functional-form test. The failure is consistent with mixing administered and market-linked pricing periods and with the unusual COVID pricing months. Restricting the preferred model to the post-2010 period and excluding April to September 2020 is thus an economic specification choice and not a cosmetic sample trim.

In Figure 1 the preferred wholesale fuel result is read through the regime lens. WPI Fuel and Power pass-through post-2010 is much larger than the pre-2010 estimate. This is consistent with a trend towards more market-linked fuel pricing, but it is not a clean causal estimate of deregulation. The Table 2 preferred excluding-COVID estimate is close to the post-2010 split in the figure and has cleaner functional-form diagnostics.



**Figure 1. Pre/post-2010 wholesale oil pass-through for headline WPI and WPI Fuel and Power. Source: Author's estimates from WPI model outputs.**

### 4.3 Retail petrol and the consumer fuel bridge

The PPAC Delhi retail petrol equation is the most directly connected fuel link to consumers in this chain. Over 245 observations with Brent as the shock variable,  $CPT^+ = 0.3459$  ( $p < 0.001$ ) and  $CPT^- = 0.1912$  ( $p = 0.0002$ ). The asymmetry p-value is 0.0999, so equality is rejected only at the ten per cent significance level. The result should therefore be viewed as indicative of greater upward than downward pass-through for retail petrol, but not

as conclusive evidence of asymmetric pricing. The retail petrol model meets the mandatory mechanism gate criteria; hence, it may carry the channel claim.

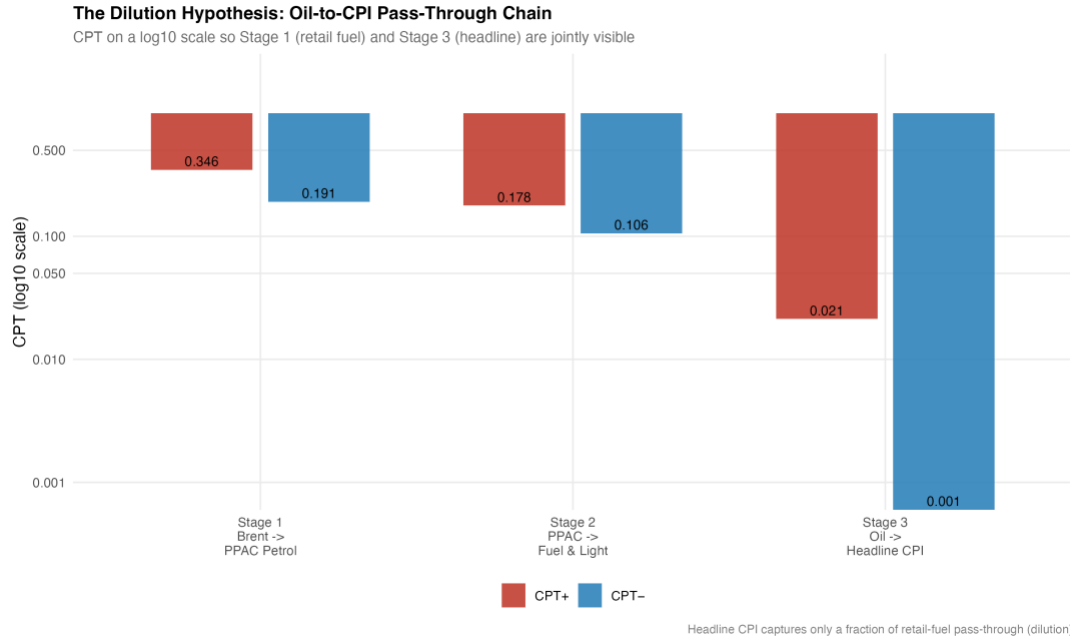
CPI Fuel and Light is the consumer fuel layer that sits between retail fuel prices and broad consumer inflation. With the PPAC petrol shock as the regressor,  $CPT^+ = 0.1777$  ( $p = 0.0021$ ) and  $CPT^- = 0.1058$  ( $p = 0.1741$ ) over 164 observations. The positive pass-through is statistically significant; the negative side is not. The result is treated as bridge evidence rather than as a primary consumer-price result, because the sample only begins in 2011-05 and is therefore considerably shorter than the headline CPI sample. The estimate is consistent with retail fuel pressure entering the fuel-sensitive consumer layer with attenuation.

The contrast between these two consumer-side layers is useful. Retail petrol is a narrow, policy-sensitive retail price and can move sharply when international prices are passed on. CPI Fuel and Light is already a basket component. It includes household fuel items and therefore smooths the retail petrol shock. The decline from 0.3459 in the retail petrol equation to 0.1777 in the bridge equation is consistent with partial transmission rather than a full one-for-one movement. Because the bridge sample is short, the estimate is not used to make a 20-year headline claim.

#### ***4.4 Headline CPI endpoint and integrated attenuation***

Headline CPI is where the oil signal becomes weak. The positive cumulative pass-through is  $CPT^+ = 0.0213$  with  $p = 0.1220$ , and the negative side is  $CPT^- = 0.0006$  with  $p = 0.9375$ . The positive coefficient has the expected sign, but it is not statistically significant at conventional levels. Asymmetry is not rejected ( $p = 0.2408$ ). The model passes the mandatory M1 diagnostic gate, so the failure to find significant pass-through is an empirical result, not an artefact of a rejected specification.

The ranking of positive pass-through estimates gives the central result, but it must be read carefully. The preferred WPI Fuel and Power estimate is numerically the largest, and retail petrol is the largest consumer-side direct fuel response. CPI Fuel and Light is smaller, while the broad headline indices are much smaller, especially headline CPI. The formal Wald test for equality of retail petrol and headline CPI positive pass-through rejects the null at  $F = 14.3499$ ,  $p = 0.0002$ . The evidence is therefore best read as attenuation across price layers rather than absence of oil-price transmission. Because not every layer shares the same sample window or shock variable, the ranking is supported most directly for the consumer chain. Figure 2 visualises that chain.



**Figure 2. Cumulative positive pass-through across the consumer-price chain. Source: Author's estimates from PPAC and CPI model outputs.**

The consumer chain is re-estimated on the overlapping consumer-side window, with small N differences caused by lag construction, in a common-sample exercise that reduces the risk of sample composition driving the ranking. Stage 1 (Brent to PPAC petrol) gives  $CPT^+ = 0.4007$  ( $p = 0.0001$ ). Stage 2 (PPAC petrol to CPI Fuel and Light) gives  $CPT^+ = 0.1777$  ( $p = 0.0021$ ). Stage 3 (rupee oil to headline CPI) gives  $CPT^+ = 0.0064$  ( $p = 0.6355$ ). The qualitative pattern is able to survive the common-sample restriction.

Predictive-precedence tests provide supporting evidence but are not used as structural causality tests. In the WPI system, lagged rupee oil changes forecast headline WPI inflation ( $F = 7.4288$ ,  $p < 0.001$ ) and WPI Fuel and Power inflation ( $F = 15.2231$ ,  $p < 0.001$ ), while the reverse WPI-to-oil direction is not supported. On the consumer side, Brent predicts PPAC petrol ( $F = 10.7127$ ,  $p < 0.001$ ). Lagged rupee oil terms predict headline CPI only at the ten per cent level ( $F = 2.2861$ ,  $p = 0.0794$ ), and PPAC petrol does not clearly predict CPI Fuel and Light in the bridge equation ( $p = 0.1126$ ). This matches the main estimates: the oil signal is easier to pick up near fuel prices than in broad CPI.

## 5. Robustness, diagnostics, and limitations

The headline WPI result is robust to decomposing the rupee oil shock into a Brent component and an exchange-rate component. The decomposition gives  $CPT^+ = 0.0309$  ( $p = 0.0234$ ) and  $CPT^- = 0.0375$  ( $p < 0.001$ ), with an asymmetry p-value of 0.7039. These figures are within rounding distance of the rupee-shock benchmark and confirm that the modest wholesale endpoint result does not depend on combining the two components. Bootstrap symmetry tests with 4,999 replications fail to reject  $CPT^+ = CPT^-$  for headline WPI and for WPI Fuel and Power, which is consistent with the HAC-based asymmetry tests in Table 2.

The diagnostics are summarised in Table 3. The headline WPI model passes the Breusch-Godfrey test for serial correlation, the HAC-RESET test for functional form, and the recursive CUSUM test for parameter stability. The preferred WPI Fuel and Power model



passes the same gate. The older full-sample pooled Fuel and Power model fails HAC-RESET and is retained only as context. The headline CPI M1 model passes the mandatory gate. The PPAC retail petrol model passes the mechanism gate. Two alternative CPI specifications explored in the supporting work, M2 and M3, are not claim-bearing because their diagnostics are rejected. The symmetry checks do not reject symmetry in the main wholesale and consumer endpoint models, so asymmetry is not the central finding of the paper. Unit-root tests support using log differences, with the WPI inflation KPSS caveat read alongside the Bai-Perron break evidence.

**Table 3. Diagnostics and robustness summary.**

| Check                         | Result                              | Interpretation                        |
|-------------------------------|-------------------------------------|---------------------------------------|
| Headline WPI diagnostics      | BG, HAC-RESET, Rec-CUSUM pass       | Main WPI model accepted               |
| Preferred WPI Fuel and Power  | BG, HAC-RESET, Rec-CUSUM pass       | Wholesale fuel mechanism accepted     |
| Pooled WPI Fuel and Power     | BG, Rec-CUSUM pass; HAC-RESET fails | Context only, not preferred inference |
| Headline CPI M1 gate          | Pass                                | Main CPI endpoint accepted            |
| PPAC mechanism gate           | Pass                                | Direct fuel channel accepted          |
| CPI M2, M3 gates              | Diagnostics fail                    | Excluded from main claims             |
| Bootstrap symmetry (WPI, CPI) | Symmetry not rejected               | Asymmetry not central                 |

*Note:* BG = Breusch-Godfrey; HAC-RESET = HAC-robust Ramsey RESET; Rec-CUSUM = recursive cumulative sum stability test. *Source:* Author's diagnostic outputs.

A pre/post-2010 split of the wholesale models offers a fourth piece of context. For headline WPI,  $CPT^+$  rises from 0.0117 ( $p = 0.3812$ ) in the pre-2010 sample to 0.0741 ( $p = 0.0043$ ) in the post-2010 sample. For WPI Fuel and Power,  $CPT^+$  rises from 0.0922 ( $p = 0.1679$ ) to 0.5241 ( $p < 0.001$ ). The post-2010 estimates are consistent with stronger pass-through around the period of more market-linked fuel pricing, including petrol deregulation in June 2010 and diesel deregulation in October 2014. The split is not a clean causal estimate of deregulation, because other macroeconomic and policy changes occurred over the same window. The preferred Fuel and Power model is the stricter post-2010 version that excludes April to September 2020 and gives  $CPT^+ = 0.5205$ . The split is reported as institutional evidence.

The limitations are stated plainly. The estimates are reduced-form short-run projections, not structural causal effects. Different layers have different sample spans, although the common-sample exercise reproduces the qualitative ranking. CPI Fuel and Light has a shorter sample than headline CPI and WPI, and is therefore used as bridge rather than primary evidence. PPAC retail petrol uses Delhi prices, which are treated as a retail fuel-channel proxy rather than a complete national retail measure. The preferred WPI Fuel and Power model uses a shorter post-2010 excluding-COVID sample; the full-sample pooled model is shown only as caveated context. The pre/post-2010 split is suggestive institutional evidence, not a clean deregulation experiment. The layered map is not a single mechanical WPI-to-CPI

causal chain; it is a set of layer-by-layer pass-through equations that share a common interpretation.

## 6. Discussion and conclusion

The result of this paper is not that oil is unimportant for Indian inflation. Retail petrol tracks Brent strongly, the wholesale fuel layer responds firmly to rupee oil shocks, and CPI Fuel and Light shows a smaller but statistically significant bridge response. The result is that headline CPI absorbs and dilutes oil shocks because the CPI basket is broad. Food, housing, services, health, and education together dominate the index, and the direct fuel weight, even augmented by indirect cost effects, is too small for the oil signal to dominate the headline figure on a monthly basis.

For inflation monitoring, the implication is narrow but useful. A reader who monitors only headline CPI may miss near-fuel price pressure, because the oil signal is small at that layer and is statistically insignificant in the present sample. Monitoring a small set of fuel-sensitive layers, such as retail petrol, WPI Fuel and Power, and CPI Fuel and Light, carries more information about short-run pass-through than the headline aggregate alone. This is consistent with earlier work on Indian oil pass-through (Mandal et al., 2012; Bhanumurthy et al., 2012; Pradeep, 2022), and adds the layered organisation as a way to read those results.

The conclusion can be put in short form. The research question is whether oil-price shocks propagate uniformly across Indian price layers or whether they become less pronounced as the index gets broader. The estimates support a descriptive attenuation reading: WPI Fuel and Power and retail petrol are the high-pass-through fuel layers, CPI Fuel and Light is smaller, headline WPI is moderate, and headline CPI is weak and statistically insignificant. The formal consumer-chain test rejects equality between retail petrol and headline CPI. The preferred WPI Fuel and Power estimate passes the diagnostic gate, while the full-sample pooled version is retained only as context. Asymmetry is not the main finding; only retail petrol shows a marginal asymmetry result at the ten per cent level. The post-2010 split supports stronger pass-through under more market-linked fuel pricing, but it is institutional rather than causal evidence.

The main lesson is that oil shocks do not disappear in India, but they lose force as they move from fuel prices to broad consumer inflation.

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